Problem 2. (Z & T, Ch. 1, Pr. 4.6 modified)

(a) (i) Find the general sol'n of the system by elim-ng one var-le.

$$\begin{cases} \frac{dx_1}{dt} = x_2 & (1) \\ \frac{dx_2}{dt} = -x_1 & (2) \end{cases}$$

(ii) Show that your sol'n is correct by substituting back to (1) & (2)

Sol'n (i) By (1), $x_2 = \frac{dx_1}{dt}$. If I substitute this into (2)

We get

$$\frac{d}{dt}\left(\frac{dx_1}{dt}\right) = -x_1 \Rightarrow \text{by (2).}$$

(3) $$\boxed{\frac{d^2x_1}{dt^2} = -x_1}$$ Solving this by the method for

2nd order constan coef-t linear ODE:

(4) $$x_1'' + x_1 = 0$$

Let's assume the Linearly independent (L.I) sol'ns of the (4) are in the form of

(5) $$x_1 = e^{rt} \Rightarrow x_1' = re^{rt} \ \& \ x_1'' = r^2e^{rt}.$$

Plugging (5) into (4) yields

$$r^2e^{rt} + e^{rt} = 0 \quad \{ : e^{rt} \neq 0 \quad \text{BS of the eq'n}$$

$$r^2 + 1 = 0 \Rightarrow r = \pm i$$

So by super position principle, general sol-n for (4) is

(6) $$x_1 = c_1\cos(t) + c_2\sin(t)$$

So, $x_1 = C_1 \cos(t) + C_2 \sin(t)$, and by (2)

(2) $$\frac{dx_2}{dt} = -x_1 \Rightarrow \frac{dx_2}{dt} = -\left[C_1 \cos(t) + C_2 \sin(t)\right]$$

(7) $$\frac{dx_2}{dt} = -C_1 \cos(t) - C_2 \sin(t),$$ solving this using sep'n of var's:

$$\int dx_2 = \int -C_1 \cos(t)\,dt - \int C_2 \sin(t)\,dt + C_3$$

(8) $$x_2 = -C_1 \sin(t) + C_2 \cos(t) + \underline{C_3}$$ (Having C_3 here ?2 But I.C.'s are 2)

Since we used (2) to find x_2 in (8), it surely satisfies (2), yet, we should check for (1) (indeed that will tell us $C_3 = ?$)

(1) $$\frac{dx_1}{dt} = x_2 \overset{\text{by (6) \& (8)}}{\Rightarrow} \frac{d}{dt}\left[C_1 \cos(t) + C_2 \sin(t)\right] = \underbrace{-C_1 \sin(t) + C_2 \cos(t) + C_3}_{x_2}$$

$$\Rightarrow \underline{-C_1 \sin(t)} + \underline{\underline{C_2 \cos(t)}} = \underline{-C_1 \sin(t)} + \underline{\underline{C_2 \cos(t)}} + C_3 \Rightarrow$$

$C_3 = 0$, so $x_2(t)$ is

(9) $$\boxed{x_2 = -C_1 \sin(t) + C_2 \cos(t)}$$ & by (6)

(6) $$\boxed{x_1 = C_1 \cos(t) + C_2 \sin(t)}$$

(ii) Substitution for verification:

(1) $$\frac{dx_1}{dt} = x_2 \Rightarrow \frac{d}{dt}\left[C_1 \cos(t) + C_2 \sin(t)\right] = -C_1 \sin(t) + C_2 \cos(t) = x_2(t) \checkmark$$

(2) $$\frac{dx_2}{dt} = -x_1 \Rightarrow \frac{d}{dt}\underbrace{\left[-C_1 \sin(t) + C_2 \cos(t)\right]}_{x_2} = -C_1 \cos(t) - C_2 \sin(t) = -x_1(t) \checkmark$$

(B) Use the ↓ I.C.'s:

$$\begin{cases} x_1(0) = 1 \\ x_2(0) = 1 \end{cases} \Rightarrow \begin{cases} C_1 \cdot \cos(0) + C_2 \sin(0) = 1 \\ -C_1 \sin(0) + C_2 \cos(0) = 1 \end{cases} \Rightarrow \begin{cases} C_1 = 1 \\ C_2 = 1 \end{cases},$$ thus by (6) & (9),

$\boxed{x_1(t) = \cos(t) + \sin(t)}$, and $\boxed{x_2(t) = -\sin(t) + \cos(t)}$.